Independent events

1 Write an equation for $P\left(A \text{ and } B\right)$ in terms of $P\left(A\right)$ and $P\left(B\right)$ given that the two events are independent.

2 A coin is tossed twice.

a Construct a tree diagram showing the results of the given experiment.

b Use the tree diagram to find the probability of getting:

i Exactly 1 head.

ii 2 heads.

iii No heads.

iv 1 head and 1 tail.

3 An archer makes two attempts to hit a target, and the probability that he hits the target on any one attempt is $\frac{1}{3}$. Find the probability that the archer will miss the target on both attempts.

4 A computer generates two random numbers between 1 and 100 (inclusive). Using the product rule, find the probability that the two numbers are:

a The same.

b Different.

c 89 and 83 in that order.

d 89 and 83 in any order.

e 14 and 36 in any order.

f Not 16 and 95.

5 A coin is tossed, a die is rolled, and a card is randomly selected from a normal deck of cards.

a Find the total number of outcomes.

b Find the probability that the coin lands on heads, the die lands on 6 and an Ace of Hearts is selected from the deck of cards.

c Find the probability that the coin lands on tails, the die lands on an even number, and the card selected is red.

6 In a game of Monopoly, rolling a double means rolling the same number on both dice. When you roll a double this allows you to have another turn. Find the probability that Sarah rolls:

a A double 1.

b A double 5.

c Any double.

d Two doubles in a row.

7 Christa enters a competition in which she guesses the 3-digit code (from 000 to 999) which cracks open a vault containing one million dollars. If the 3-digit number to open the vault is randomly generated by a computer, find the probability that it is:

a An odd number.

b An even number (including 000).

c A number greater than 123.

d A number divisible by 10.

e A number less than 321.

8 A fair die is rolled twice. Find the probability of:

a Rolling a 6 and a 1 in any order.

b Rolling a 6 and then a 1.

9 A pile of playing cards has 4 diamonds and 3 hearts. A second pile has 2 diamonds and 5 hearts. One card is selected at random from the first pile, then the second.

a Construct a probability tree of this situation with the correct probability on each branch.

b Find the probability of selecting two hearts.

10 Students were asked what they are allergic to. The table shows the results:

	Allergic to nuts	Not allergic to nuts
Allergic to dairy	14	12
Not allergic to dairy	25	25

If a student is chosen at random, find the probability that:

a The student is allergic to dairy.

b The student has an allergy.

c The student does not have an allergy.

d The student is also allergic to dairy if he or she is allergic to nuts.

Dependent events

11 Mae deals two cards from a normal deck of cards. Calculate the probability that she deals:

- **a** Two 10s
- **b** Two red cards
- **c** Two diamonds
- **d** A 10 of spades and an Ace of diamonds, in that order
- **e** A 10 of clubs and an Ace of spades, in any order

12 A carton contains a dozen eggs, of which 4 contain no yolk. If 3 eggs are chosen at random for a cake, find the probability that they all have no yolk.

13 From a set of 10 cards numbered 1 to 10, two cards are drawn at random without replacement. Find the probability that:

- **a** Both numbers are even.
- **b** One is even and one is odd.
- **c** The sum of the two numbers is 12.

With or without replacement

14 Eileen randomly selects two cards, with replacement, from a normal deck of cards. Find the probability that:

- **a** The first card is a queen of spades and the second card is a 4 of clubs.
- **b** The first card is spades and the second card is a 4.
- **c** The first card is a Queen and the second card is black.
- **d** The first card is not a 7 and the second card is not Clubs.

15 Beth randomly selects three cards, with replacement, from a normal deck of cards. Find the probability that:

- **a** The cards are queen of diamonds, king of spades, and king of diamonds, in that order.
- **b** The cards are all black.
- **c** The first card is a 9, the second card is a heart and the third card is red.
- **d** The cards are all hearts.

16 A chess player is placed into a draw where in each match he has a 30% chance of winning. Find the probability that:

a He wins his first two matches.

b He wins his first three matches.

c He wins his first two matches and then loses his third match.

17 Amelia randomly selects two cards, with replacement, from a normal deck of cards. Find the probability that:

a Both cards are red

b Both cards are the same color.

c Both cards are of different colors.

18 A bag contains the letters A, B, C, D and E. Two letters are randomly drawn out of the bag without replacement.

a Construct a probabilty tree showing all possible outcomes.

b Find the probability of the following:

i A is drawn followed by B.

ii Two A's are drawn.

iii C is not drawn.

iv D is drawn.

v E is drawn and B is not.

vi C or D is drawn.

vii Neither C nor D are drawn.